

EXCEL BUSINESS ANALYTICS / PROFESSIONAL PORTFOLIO

NorthStar Retail Analytics

Business Performance, Profitability & Customer Insights

An independent Excel analytics case study using a fictional retail business and a synthetic 2025 transaction dataset.

SCOPE	TOOLS & OUTPUT
2,000 retail transactions	Microsoft Excel · PivotTables · formulas · KPI analysis
2025 analytical period	Three reference dashboards
Five source tables	Products · Stores · Salespeople · Customers · Transactions

Portfolio purpose: Demonstrate business problem framing, data validation, KPI development, dashboard communication, and actionable recommendations.

Growth without equivalent margin improvement

NorthStar Retail is a fictional multi-region retailer. The analysis examines 2025 sales performance and compares selected headline metrics with supplied 2024 scenario benchmarks. The central management question: why is revenue growing faster than gross profit?

PERFORMANCE MEASURE	2024 BENCHMARK	2025 ANALYSIS
Revenue	$1,115,400	$1,204,636
Gross profit	$345,305	$357,391
Gross profit margin	30.96%	29.67%
Revenue growth	—	+8.0%
Gross profit growth	—	+3.5%

The headline finding

Revenue increased by approximately 8.0% while gross profit increased by only 3.5%. Gross profit margin fell by about 1.29 percentage points. These figures motivate a closer look at product mix, discounting, regional differences, and sales channels.

Decision questions

- Product mix — Which categories drive sales, and which contribute stronger margins?
- Regional performance — Are high-revenue regions also the most profitable?
- Commercial controls — Are any transactions being sold below recorded cost?

Evidence note: 2024 values are supplied comparison benchmarks in the workbook, not results recalculated from a 2024 transaction table. The 2025 results reflect the project reference analysis.

A reproducible Excel analysis workflow

The project workbook combines a fictional retail scenario with structured source tables, analytical worksheets, reference calculations, and three dashboards. The analysis uses 2,000 transactions from January through December 2025.

DATA TABLE	ANALYTICAL PURPOSE
Transactions	Sales-level records used to calculate revenue, costs, gross profit, units, and performance over time.
Products	Product identifiers and categories for product mix and profitability.
Stores	Store and region attributes for geographic comparisons.
Salespeople	Salesperson attributes for sales performance analysis.
Customers	Customer attributes for segment and channel-oriented analysis.

Preparation & quality controls

- Relationship checks — Confirm transaction IDs and lookup keys map to the correct reference tables.
- Type and formula checks — Review currency values, numeric calculations, date ranges, and category mappings.
- Reconciliation — Compare transaction-derived totals with dashboard and pivot-table KPI outputs.
- Exception review — Inspect negative gross profit transactions separately rather than removing them without justification.

Core metric definitions

Revenue = sum of recorded transaction sales; Gross profit = revenue minus recorded cost; Gross profit margin = gross profit ÷ revenue. These are gross, not net, profitability measures.

Scope limitation: the workbook is a simulated educational dataset. Findings demonstrate analytical reasoning; they do not establish actual market outcomes or customer behavior.

Sales growth and profitability pressure

The Executive Dashboard provides a management-level overview of revenue, gross profit, margin, and changes over time. It highlights the importance of viewing growth and profitability together.

2025 KPI	RESULT
Total revenue	$1,204,636.05
Gross profit	$357,391.05
Gross profit margin	29.67%
Transactions	2,000
Units sold	3,083

Monthly pattern

- December — Highest revenue month: approximately $152,183.
- July — Lowest revenue month: approximately $70,752, but the highest monthly gross profit margin at 32.90%.
- November — Lowest monthly gross profit margin at 26.16%.

Interpretation

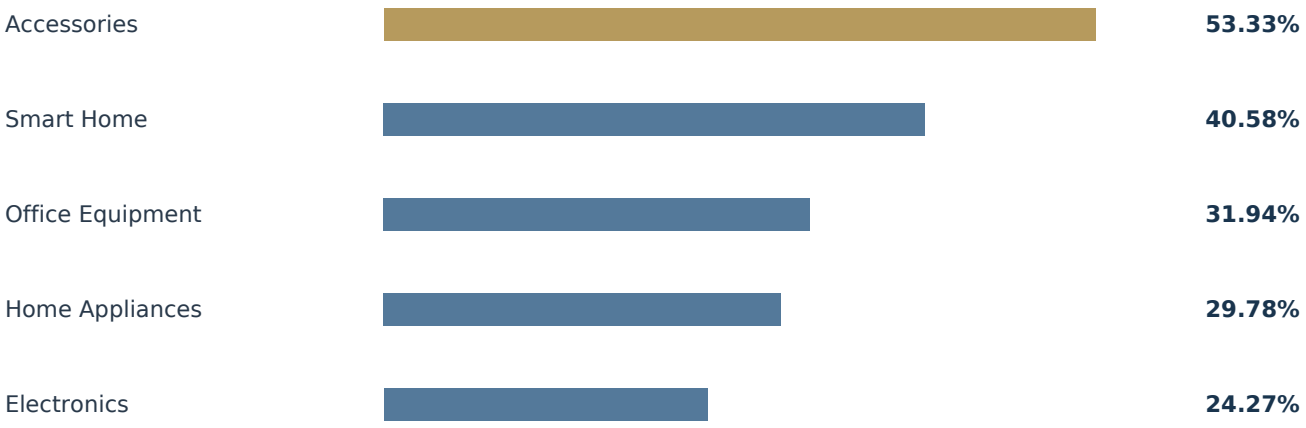
Higher sales volume does not automatically translate to stronger margin performance. The contrast between July and November suggests management should review category mix, transaction discounts, and unit economics by month before changing pricing or promotional policy.

Portfolio skill demonstrated: translating financial KPIs into an executive narrative, rather than treating each metric as an isolated chart.

Revenue concentration vs. margin quality

Electronics generated approximately $589,845, nearly 49% of total revenue, but had a lower gross margin than the business overall. Accessories had a much higher margin on a smaller revenue base.

Gross profit margin by category



Key product results

INDICATOR	PRODUCT / RESULT
Highest revenue	15-inch Laptop · $199,509.10
Highest gross profit	15-inch Laptop · $47,189.10
Most units sold	Wireless Mouse · 241 units

Business implication: test whether accessory attachment, bundling, or product mix improvements could lift gross margin without assuming that a high-margin category can automatically scale.

Findings that suggest targeted investigation

Regional comparisons reveal that sales leadership and margin leadership are not identical.

REGION	REVENUE	GROSS MARGIN
Ontario	$435,335	29.22%
Quebec	$288,009	29.71%
British Columbia	$200,777	29.04%
Alberta	$177,958	30.46%
Atlantic Canada	$102,556	31.32%

Customer and channel highlights

- Consumer segment — Approximately $689,324 in revenue, or 57.2% of the total; 30.20% gross margin.
- In-store channel — Approximately $664,861 in revenue, or 55.2% of total; 29.99% gross margin versus 29.28% online.

Discount exceptions

Four transactions were identified with negative gross profit in the reviewed analysis: TRX0628 (−$21.40), TRX0633 (−$10.70), TRX0730 (−$62.10), and TRX0845 (−$10.70). These merit investigation, but their small number does not establish the primary cause of the overall margin decline.

Interpretation: differences may reflect product mix, pricing, discounting, or other recorded factors. The dataset supports descriptive comparison, not causal attribution.

From descriptive insights to management action

PRIORITY	RECOMMENDED ACTION
1 · Margin visibility	Track revenue, gross profit, and margin together by month, category, and product.
2 · Discount controls	Review transactions with negative gross profit and assess warnings for below-cost pricing.
3 · Product mix	Test accessory attachment and complementary product bundles using measured pilot results.
4 · Regional learning	Investigate pricing and product mix differences between Ontario and higher-margin regions.
5 · Ongoing reporting	Use consistent KPI definitions and periodic data-quality checks to sustain dashboard reliability.

Skills demonstrated

Excel analytics; PivotTables; lookup and calculation formulas; structured data validation; relational thinking; KPI definition; dashboard design; business communication; analytical documentation; and end-to-end educational product development.

Analytical limitations

The retailer and transactions are synthetic. 2024 comparison values are supplied scenario benchmarks. The analysis reports gross profit rather than net income; no verified operating costs, controlled experiments, or post-recommendation outcomes are available. Recommendations are proposals for investigation, not demonstrated improvements.

Project context

NorthStar Retail Analytics was developed as the first Insight Foundry guided learning product. This separate case study highlights the underlying analytical decisions and portfolio-relevant skills.

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